

Total No. of Questions : 10]

SEAT No. :

P19

[Total No. of Pages : 2

[5871]-519

B.E. (Civil)

AIR POLLUTION AND CONTROL
(2015 Pattern) (Semester - II) (Elective-III)

Time : 2 ½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Figures to the right side indicate full marks.
- 3) Draw neat figures wherever necessary.
- 4) Assume any missing data if necessary.
- 5) Use of scientific calculators is allowed.

Q1) a) Explain super adiabatic atmospheric condition. [5]

- b) The emission of SO₂ from a coal fired power plant, at a rate of 1500 g/s. The wind speed is 4 m/s on a sunny afternoon. Determine the effective stack height. Stack parameters : Height=130 m, Diameter = 1.5m, Exit velocity = 12 m/s, Temperature =320 °C, Atmospheric condition, P=100 kPa, T=25 °C. [5]

OR

Q2) a) Describe method of ambient air quality analysis for suspended particulate matter. [5]

- b) What are the devices used for sampling of gases and vapours? Describe any one in detail. [5]

Q3) a) Explain sick building syndrome. [5]

- b) Explain physical and chemical changes in indoor air quality. [5]

OR

Q4) a) Explain air cleaning system. [5]

- b) Write National Ambient Air Quality Standards 2009. [5]

P.T.O.

- Q5) a)** With neat sketch explain the principle, construction and working of cyclone. **[8]**
- b)** Name and describe three control devices developed in control of automotive emission. **[8]**

OR

- Q6) a)** A horizontal parallel plate electrostatic precipitator consisting of three gas passages $4.8 \text{ m} \times 4.8 \text{ m}$ deep with 30 cm plate spacing is to treat $540 \text{ m}^3/\text{min}$ of air. Due to a rapid emission before inlet to the precipitator, the velocity profile is not fully developed. As a result, the volume rate of flow in the middle duct is twice that of the outer duct. Calculate the collection efficiency. Use migration velocity of 0.8 m/s . **[8]**
- b)** Explain the principles of scrubber with a neat sketch. What are the advantages and disadvantages of scrubber? **[8]**
- Q7) a)** Discuss the application of land use planning as a method of air pollution control. **[8]**
- b)** Explain the economics of air pollution control on the basis of cost/benefit ratio and optimization **[8]**

OR

- Q8) a)** Explain how air pollution due to automobile are controlled. **[8]**
- b)** What is land use planning and write its importance. **[8]**
- Q9) a)** Write roles of public and impact assessment agency in the EIA process. **[6]**
- b)** Write short note on EIA by Matrix method. **[6]**
- c)** Explain the EIA cycle and procedures. **[6]**

OR

- Q10) a)** Write the environmental rules 1999 (siting of industries) as per the notification of Ministry of Environment and Forest. **[9]**
- b)** Explain roles of different authorities in the EIA process. **[9]**

